

Cerebellar structural network differentiates schizophrenia from bipolar disorder and controls



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Introduction

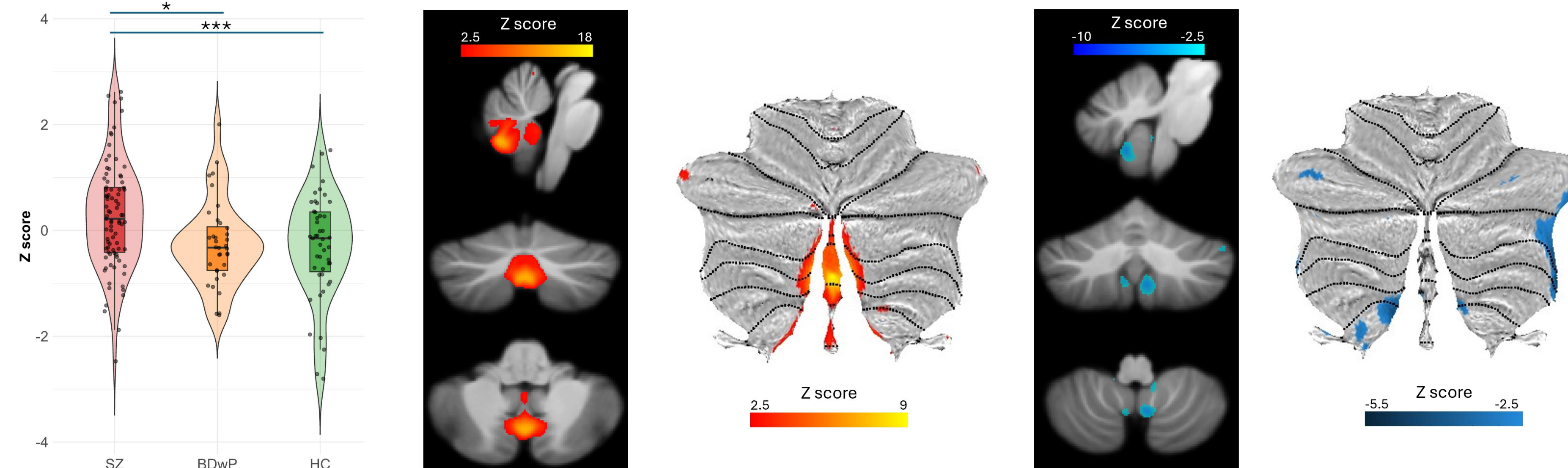
- Schizophrenia research has primarily focused on cortical and subcortical brain regions
- The cerebellum remains relatively understudied, despite its critical roles in cognition, emotion, and motor coordination
- Source-based morphometry (SBM) applies independent component analysis (ICA) to gray matter data
- SBM identifies spatially distinct structural patterns that covary across individuals
- This multivariate approach captures network-level anatomical organization beyond traditional univariate methods (e.g., voxel-based morphometry)

Methodology

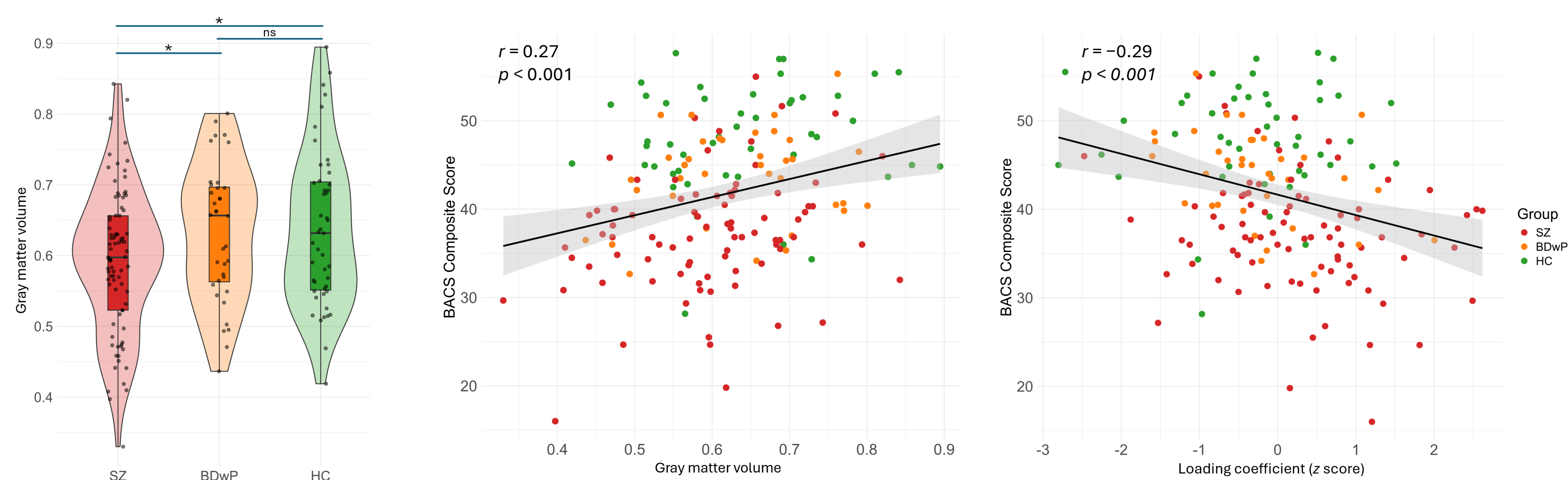
- Data from the Psychosis Human Connectome Project (P-HCP), University of Minnesota
- Sample: 168 participants (Schizophrenia: n = 85, Bipolar disorder with psychotic features: n = 36, Healthy controls: n = 47)
- T1-weighted images preprocessed using the ENIGMA Cerebellum Volumetrics Pipeline
- Cerebellum isolated using ACAPULCO
- Cerebellum normalized to standard space and segmented into gray and white matter using SUIT
- SBM performed using the SBM module of the GIFT Toolbox
- Group comparisons conducted using general linear models controlling for age, sex, and frame-wise displacement

Results

- One independent component (IC) showed a significant diagnostic group effect ($p < 0.05$, Bonferroni-corrected)
- This IC differentiated schizophrenia (SZ) from both healthy controls (HC) and bipolar disorder with psychotic features (BDwP)
- Positive loadings: vermis VIIIa, bilateral Crus I, right lobule IX
- Negative loadings: bilateral Crus I, bilateral lobule I



- Voxel-based morphometry showed reduced gray-matter volume in negatively loaded regions in schizophrenia (SZ)
- Cognitive performance was significantly associated with:
 - Gray matter volume
 - Structural network loading coefficients



- Mediation analyses revealed:
 - A strong direct effect of diagnostic group on IC loadings (effect size = -0.42)
 - Small, nonsignificant indirect effects via cognition

Conclusion

- Cerebellar gray matter structural networks are altered in schizophrenia relative to healthy controls and, in some networks, bipolar disorder with psychotic features
- SBM identified spatially distinct cerebellar networks that may be missed by traditional univariate approaches
- Key alterations involved bilateral Crus I and bilateral lobule IX, underscoring cerebellar roles beyond motor function
- Findings support the cerebellum as a critical node in neural circuits underlying schizophrenia
- Integrating structural and functional data may further clarify cerebellar contributions to cognitive and clinical features
- Results highlight the importance of including cerebellar analyses in neuroimaging studies of psychotic disorders

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